
Truncated automatic sparse differentiation for machine learning interatomic potentials

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Abstract

Machine learning interatomic potentials (MLIPs) learn the mapping from atomic positions to potential energy. The forces, the negative gradient of this energy, drive molecular dynamics and are readily obtained using automatic differentiation. Higher-order derivatives, most notably the Hessian, describe collective motion and allow the direct prediction of experimental observables, but are considered computationally inaccessible for large systems. We suggest a solution: In physical systems, interactions decay with distance, and most MLIPs build on this locality through message passing up to a finite receptive field. This implies both sparsity of higher-order derivatives and their decay with distance. This structure can be exploited using automatic sparse differentiation (ASD). We explain how to compute the sparsity pattern for MLIP derivatives and demonstrate that, for multiple foundation MLIPs, ASD computes full Hessians of large porous materials exactly, but with modest speedups at best. The larger gains come from truncated ASD: discarding small, but nonzero, Hessian entries between distant atoms yields order-of-magnitude speedups with negligible impact on predicted observables.

1 Introduction

Machine learning interatomic potentials (MLIPs), i.e., learned approximations to the Born and Oppenheimer [1927] potential energy surface (PES) based on first-principles reference data, have become increasingly valuable tools for computational materials science, chemistry, and biology [Unke et al., 2021, Deringer et al., 2019]. In practice, not only the potential energy, but also its derivatives carry essential information for atomistic modeling. First derivatives describe forces acting on atoms and allow the simulation of atomic motion through molecular dynamics. Higher-order derivatives describe the curvature of the PES. Under the assumption of small deviations around a local minimum (the *harmonic approximation*), the Hessian gives rise to collective modes of motion (the *phonons*), a long-established model of vibrations in molecules and solids that underpins the prediction of experimental observables such as heat capacity, vibrational spectra, and thermal transport [Debye, 1912, Dove, 1993]. Third and higher-order derivatives describe the *anharmonic* contributions and are interpreted as scattering between phonon modes, determining their finite lifetimes and linewidths.

Obtaining derivatives of MLIPs is therefore an important task. Since the total energy is a scalar, automatic differentiation (AD) allows the calculation of forces (its negative gradient) in a single backward pass, at the same asymptotic cost as the energy prediction itself. This is not the case for higher-order derivatives: if the forward pass is $\mathcal{O}(N)$, as for the MLIPs considered here, the full Hessian takes $3N$ derivative passes of $\mathcal{O}(N)$ each, hence $\mathcal{O}(N^2)$; the full third-derivative tensor $(3N)^2$ passes, hence $\mathcal{O}(N^3)$, and so on. This bottleneck has restricted the use of MLIPs for vibrational analysis to modest system sizes thus far [Gönnheimer et al., 2025, Elena et al., 2025, Loew et al., 2025].

Fortunately, this asymptotic scaling is a worst-case estimate for *generic* higher-order derivatives. If derivatives have additional structure—in particular, sparsity—they can be computed more efficiently by omitting known zeros. This technique is called automatic sparse differentiation (ASD) [Curtis et al., 1974, Powell and Toint, 1979]. This work gives a blueprint for using ASD with MLIPs based on message-passing neural networks (MPNNs). We show how to compute the sparsity pattern for a given MLIP in closed form, and argue that this procedure also gives rise to a hierarchy of truncated k -hop sparsity patterns that converges to the exact sparsity pattern of the model. For common MLIPs (MACE, PET) and porous materials—metal–organic frameworks (MOFs), covalent organic frameworks (COFs), and zeolites—we show empirically that (a) ASD with the exact pattern recovers dense Hessians to floating-point accuracy at modest speedups, larger for models with smaller receptive fields, (b) Hessian entries decay rapidly with hop distance, at a rate set by the chemistry of the system, and (c) truncating the pattern at a low hop count therefore trades a controlled approximation for much larger speedups, with often negligible impact on predicted observables. Our results bring experimental observables of large systems within routine reach, opening the door to systematic evaluation of MLIPs on, and fine-tuning with, real-world measurement results.

2 Background

Machine learning interatomic potentials Under the Born and Oppenheimer [1927] approximation, the N nuclei of a molecule or material move on a PES $E = E(\{\mathbf{r}_i, Z_i\}_{i=1}^N)$ that depends only on their positions $\mathbf{r}_i$ and atomic numbers Z_i . The forces driving their dynamics are derivatives of this energy, $\mathbf{F}_i = -\partial E / \partial \mathbf{r}_i$. In tandem with the increasing availability of quantum-mechanical reference data, MLIPs [Behler and Parrinello, 2007] have emerged as a data-driven approximation to this surface. Modern MLIPs are, almost without exception, graph neural networks (GNNs) [Battaglia et al., 2018] acting on a geometric graph of atoms whose edges encode interatomic vectors within a fixed cutoff radius r_c , with each atomic energy contribution E_i local to a receptive field that grows linearly with the number of message-passing iterations. The total energy is a sum of N such contributions, each with cost bounded by the finite cutoff, so one forward pass is $\mathcal{O}(N)$.¹ Recently, a line of *foundation* MLIPs has been trained on broad slices of the periodic table and is able to predict energies and forces for arbitrary chemistries in a zero-shot setting [Batatia et al., 2025, Mazitov et al., 2025, Wood et al., 2025, Yang et al., 2024, Rhodes et al., 2025, Malosso et al., 2026].

Phonons Materials are modeled as periodic systems: only the atoms of the *unit cell*, which is tiled in space, move independently; a *supercell* combines several unit cells into a larger one to increase the number of independent degrees of freedom. The *interatomic force constants* $\Phi_{i\alpha,j\beta} = \partial^2 E / \partial r_{i\alpha} \partial r_{j\beta}$, the second-order expansion coefficients of the PES around a minimum, are simply the Hessian of the energy with respect to Cartesian positions. We write $\|\Phi_{ij}\|$ for the Frobenius norm of the 3×3 block of atom pair (i, j) . Mass-weighting yields the dynamical matrix, whose eigendecomposition gives the squared phonon frequencies ω_s^2 . The simulation cell must be large enough to contain all interactions between atoms; for MLIPs, the required size can be determined exactly from the model’s interaction range (Appendix A). Harmonic observables are thermal averages over the phonon spectrum [Dove, 1993]; we consider the isochoric heat capacity

$$C_V(T) = k_B \sum_s \frac{x_s^2 e^{x_s}}{(e^{x_s} - 1)^2}, \quad x_s = \frac{\hbar \omega_s}{k_B T}. \quad (1)$$

3 Related work

Automatic differentiation for Hessians Gönheimer et al. [2025] computed the Hessian of MACE with AD, predicting zero-shot heat capacities on Moosavi et al. [2022]’s porous-materials benchmark. Using dense AD, they report a ceiling of ~ 1900 atoms on an A100 GPU. We tackle this bottleneck.

Approximating Hessians Hessians can be used as a *training* signal for MLIPs: Rodriguez et al. [2025] showed that training on full density functional theory (DFT) Hessians improves transition-state and vibrational-spectrum prediction, while PFT [Koker et al., 2026], PHL [Rodriguez et al., 2026],

¹The relevant scaling for MLIPs is $N \rightarrow \infty$ at constant density (the *thermodynamic limit*), so the work per atomic environment stays constant.

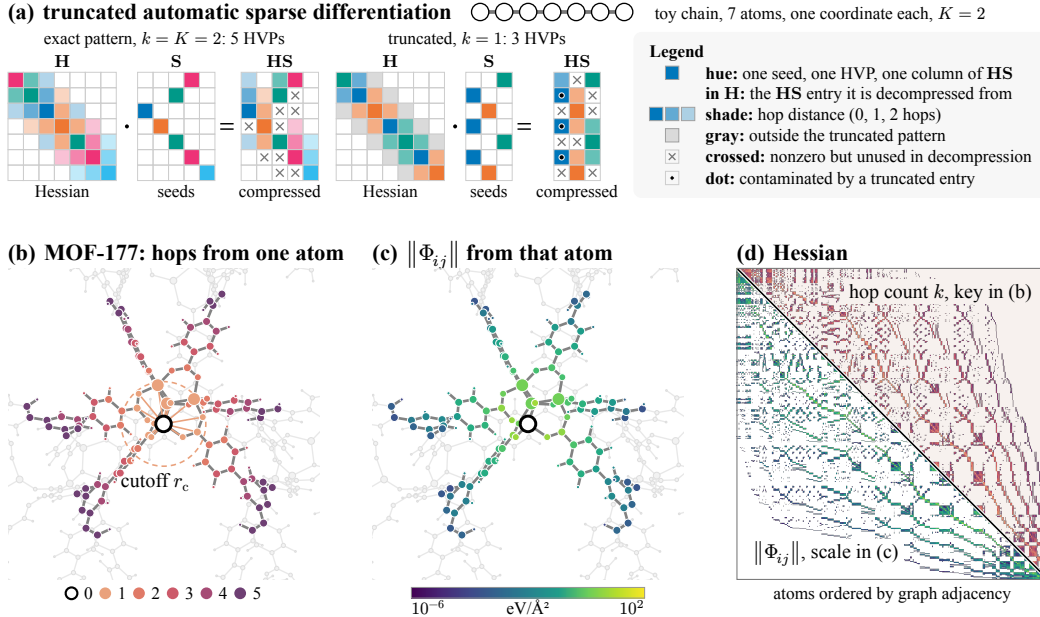


Figure 1: Truncated ASD. (a) Toy problem: a chain with 7 atoms and a model with Hessian reach of $K = 2$ hops. The Hessian $\mathbf{H}$ is decompressed from HVPs with the columns of a seed matrix $\mathbf{S}$, which yield the compressed product $\mathbf{HS}$. Since $\mathbf{H}$ is sparse, star coloring yields 5 colors (and thus 5 HVPs), fewer than the 7 of a dense evaluation. Every entry of $\mathbf{H}$ is decompressed from one entry of $\mathbf{HS}$, shown by its hue, either directly or through its symmetric partner; entries of $\mathbf{HS}$ that are sums of several Hessian entries are never read (crossed). Truncating the sparsity pattern to one hop needs only three HVPs, but the neglected two-hop couplings (gray) now contaminate entries used for decompression (dots), leading to errors (Appendix H). (b) MOF-177 with PET-XS: the atoms within $K = 5$ hops of one Zn atom on the model’s input graph. We indicate the atom’s cutoff sphere and one-hop edges, the rest of the cell is drawn in gray. (c) The same atoms colored by the force-constant block norm $\|\Phi_{ij}\|$ between each atom and the marked one, showing its decay with distance. (d) The whole Hessian of MOF-177 with one cell per pair of atoms, ordered so that graph neighbors are adjacent. Hop count above the diagonal, $\|\Phi_{ij}\|$ below, structural zeros white.

and the HORM dataset [Cui et al., 2026] make this tractable via stochastic sampling of Hessian columns or Hutchinson-style random Hessian-vector products (HVPs).

Predicting Hessians with MLIPs HIP [Burger et al., 2025] bypasses AD, directly predicting Hessians from SE(3)-equivariant features, with 10–74 $\times$ speedups over AD on small molecules; the predicted Hessians are not guaranteed consistent with the gradient of the model’s own forces. Inconsistencies of this kind can lead to problems in physical simulations [Bigi et al., 2025].

Exploiting sparsity and decay Truncating force constants by distance is routine in lattice dynamics, whether through the supercell of a finite-displacement calculation [Togo, 2023] or explicit cutoffs in force-constant fitting [Esfarjani and Stokes, 2008, Eriksson et al., 2019]. Probing methods in numerical linear algebra similarly reconstruct matrices with known sparsity or decay from few matrix-vector products [Frommer et al., 2021, Amsel et al., 2026].

4 Methods

Hessians via AD Given a function implemented as a computer program, AD automatically generates programs that evaluate Jacobian-vector products (JVPs) or vector-Jacobian products (VJPs) at the same asymptotic cost as the function itself [Griewank and Walther, 2008]. For an MLIP energy $E : \mathbb{R}^{3N} \rightarrow \mathbb{R}$, one VJP therefore yields the entire gradient, and with it all forces, at the cost of one energy evaluation (the *cheap gradient principle*, Wolfe, 1982, Baur and Strassen, 1983).

Differentiating that program again yields an HVP $\mathbf{H}\mathbf{v}$, again at the cost of a small multiple of E [Pearlmutter, 1994].² Individual products are cheap, but materializing $\mathbf{H}$ is not: each HVP $\mathbf{H}\mathbf{e}_i$ with a standard basis vector recovers one column, so the full Hessian costs $3N$ passes; at $\mathcal{O}(N)$ per pass for a linear-scaling MLIP, this amounts to $\mathcal{O}(N^2)$. Phonon observables require the full eigenspectrum, and hence the materialized Hessian—matrix-free approaches cannot be used.

Automatic sparse differentiation ASD via *compressed evaluation* [Curtis et al., 1974, Powell and Toint, 1979], recently re-popularized in the context of machine learning [Hill et al., 2025, Hill and Dalle, 2025], reduces the $3N$ HVP passes to the number c of *symmetrically orthogonal partitions* of the Hessian sparsity pattern, provided the pattern is known ahead of time. Each pass evaluates one HVP with the sum of a partition’s basis vectors, a column of the seed matrix $\mathbf{S}$, returning the sum of the corresponding columns, a column of $\mathbf{H}\mathbf{S}$; orthogonality lets individual entries be read off that sum (Figure 1a). The partitioning is found by graph coloring, specifically *star coloring* [Coleman and Moré, 1984, Gebremedhin et al., 2005, 2009], so we call c the number of *colors*; for very sparse patterns, $c \ll 3N$. The sparsity pattern need not be exact: a superset of the true one is safe and merely costs extra colors, while a subset breaks orthogonality and makes the recovery lossy. This tradeoff is the basis of the truncated ASD discussed below. We use `asdex` [Hill and Dalle, 2026] for coloring and decompression. Coloring in this particular case can be sped up by observing that the sparsity pattern is made up of 3×3 blocks, one per pair of atoms, spanning the three components of each atom’s position, with the relevant sparsity fully determined by atom-atom connectivity. We therefore only color the graph of atoms, and then assign the three coordinates of each atom three distinct colors derived from the atom’s color. This yields a valid star coloring of the full sparsity pattern at a fraction of the cost (see Appendix D).

Sparsity detection for MLIPs Using ASD requires knowing the sparsity pattern of the target derivative matrix without evaluating it. For MLIPs based on GNNs, the sparsity pattern follows directly from the input graph. Manual or empirical inspection of the model determines the receptive field of its energy terms: each term depends on the atoms within L hops of the atom (MACE) or edge (PET) it is predicted for, with L the number of message-passing layers. Since the second derivative $\partial^2 E / \partial \mathbf{r}_i \partial \mathbf{r}_j$ couples two atoms only when both lie in the receptive field of a shared energy term, the Hessian’s sparsity pattern extends at most to the diameter of that field: $K = 2L$ hops for MACE and $K = 2L + 1$ for PET; both counts are derived and verified in Appendix C. The K -hop sparsity pattern is then given by the nonzero entries of

$$\mathbf{A}^{(K)} = \sum_{k=0}^K \mathbf{A}^k, \quad (2)$$

evaluated in Boolean arithmetic, where $\mathbf{A}$ is the adjacency matrix of the input graph, $\mathbf{A}_{ij} = 1$ if atoms i and j share an edge (Figure 1b–d). This pattern is *exact*: entries outside it are structurally zero. It is also the graph the coloring acts on.

Truncated automatic sparse differentiation Truncating Equation (2) at $k < K$ yields a nested family of increasingly sparse patterns that retain couplings only up to k hops on the input graph. Truncated patterns reduce the cost of ASD, since sparser patterns admit colorings with fewer colors, and therefore require fewer HVPs and less storage. The resulting Hessian is approximate in two ways: couplings beyond k hops are discarded, and, since star coloring guarantees collision-free recovery only for a conservative sparsity pattern, the neglected couplings contaminate retained entries sharing a color. Both errors are controlled by the magnitude of the neglected couplings. Because message-passing MLIPs build up longer-ranged interactions by iterating local ones, these couplings should decay with graph distance, mirroring the decay of the underlying PES. Whether they decay fast enough for truncated Hessians to remain useful is an empirical question, answered in Section 5.

5 Experiments

Setup We evaluate three foundation MLIPs: MACE (MACE-MP-0 medium, $L=2$, $K=4$), PET-XS ($L=2$, $K=5$), and PET-S ($L=3$, $K=7$); details are given in Appendix B. We study porous materials (MOFs, COFs, zeolites) from Moosavi et al. [2022]’s benchmark: a stratified twelve-structure subset

²We refer to Dagr  ou et al. [2024] for a detailed comparison of HVP modes in JAX and PyTorch.

Table 1: Dense, truncated ($k < K$), and exact ($k = K$, shaded) Hessians of the giant MOFs. N is the number of atoms in the converged supercell. Speedups are relative to dense wall time, all end to end, including sparsity pattern construction and coloring. δC_V is the deviation of C_V at 300 K from the dense reference in per mille, with the acoustic sum rule enforced. MACE has $K=4$.

		dense		Speedup over dense					δC_V (‰)				
				truncated				exact	truncated				exact
Structure	Model	N	(s)	$k=1$	$k=2$	$k=3$	$k=4$	K	$k=1$	$k=2$	$k=3$	$k=4$	K
MIL-101	MACE	3604	3529	36.5	13.3	5.9	–	2.7	–20.77	–0.18	<0.01	–	<0.01
	PET-XS	3604	316	8.9	8.5	7.3	6.3	5.3	–109.50	–15.09	<0.01	<0.01	<0.01
	PET-S	3604	1248	23.9	16.6	11.1	7.5	1.9	–55.58	–0.22	<0.01	<0.01	<0.01
MIL-100	MACE	2788	1246	18.6	7.5	3.1	–	1.3	–7.71	<0.01	<0.01	–	<0.01
	PET-XS	2788	227	6.3	5.9	5.2	4.4	3.6	–87.99	–5.60	<0.01	<0.01	<0.01
	PET-S	2788	876	16.8	11.5	7.7	5.0	1.2	–24.81	0.03	<0.01	<0.01	<0.01
MOF-210	MACE	1854	555	10.9	8.0	5.1	–	2.6	–32.40	–0.39	<0.01	–	<0.01
	PET-XS	1854	111	3.3	3.2	3.0	3.0	2.7	–127.99	–51.02	–8.55	<0.01	<0.01
	PET-S	1854	379	8.0	6.6	5.4	4.3	1.5	–76.44	–14.13	<0.01	<0.01	<0.01
MOF-177	MACE	6464	9431	80.5	30.2	14.3	–	5.6	–26.28	<0.01	<0.01	–	<0.01
	PET-XS	808	48	1.5	1.4	1.4	1.4	1.3	–126.79	–39.20	–7.21	<0.01	<0.01
	PET-S	6464	3900	67.1	46.4	23.4	15.4	3.2	–66.16	–7.37	<0.01	<0.01	<0.01

in supercells converged with respect to each model’s interaction range (both described in Appendix A); for validation against prior work, we recompute all 233 structures in their unit cells with MACE (Appendix E). We additionally consider giant MOF unit cells: MIL-101 (3604 atoms), one of the largest unit cells among common MOFs, as well as MIL-100 (2788 atoms), MOF-210 (1854 atoms), and MOF-177 (808 atoms). All Hessians are computed in single precision (a choice ablated in Appendix E) on a single H100 GPU; heat capacities follow the pipeline described there.

Hessian entries decay with graph distance We first examine the premise of truncation: Figure 3 shows the magnitude of the force-constant blocks $\|\Phi_{ij}\|$ as a function of the hops k between atoms i and j . Entries decay by roughly two orders of magnitude per hop for MACE and slightly more than one for the PET models, so that at $k=3$ the median block norm lies 3.5 to 7 orders of magnitude below the on-site blocks. The speed of decay depends on chemistry: MOFs and COFs decay at a similar rate, while zeolites display a slower decay.

Automatic sparse differentiation is exact Next, we use both dense and sparse AD to compute the exact Hessians for two kinds of systems: the giant MOFs, for most of which no supercells are required to obtain converged vibrational properties, and smaller systems, which must be tiled up to each model’s effective interaction range. Cost (measured end-to-end) and accuracy for the giant MOFs are listed in Table 1; results for all structures in Table 4. In all cases, ASD reproduces the dense predictions to within 5×10^{-4} in relative Frobenius norm, and typically to 10^{-6} , close to expectation in single precision. However, it offers modest speedups at best: median end-to-end speedups of $1.7\times$ for MACE, $1.4\times$ for PET-XS and $1.2\times$ for PET-S, never more than $6\times$ anywhere, and barely above break-even for the densest structures. The reason for this is a lack of usable sparsity: the full interaction range of the MLIPs used in this work is comparable to even the large MOF unit cells, and the supercells used for smaller systems are constructed to have exactly the minimum size to contain that interaction radius. The only additional source of sparsity is therefore porosity, i.e., regions where the input graph is genuinely disconnected, which varies strongly between systems. Low sparsity means many colors, so there is little to gain from ASD. Predicted heat capacities are compared with published calorimetry in Appendix I, where all models overestimate the measured values.

Truncated automatic sparse differentiation trades accuracy for speed Finally, we measure Hessian error, C_V error, and speedup of truncated sparsity patterns against the dense reference, at every hop count k across the benchmark subset and the giant MOFs (Figure 2). As the sparsity pattern becomes more and more filled in with increasing k , errors and speedups decrease. Derived observables converge abruptly: at unconverged rungs the relative error in C_V is comparable to that of the Hessian, and within a single hop it drops to two to three orders of magnitude below it.

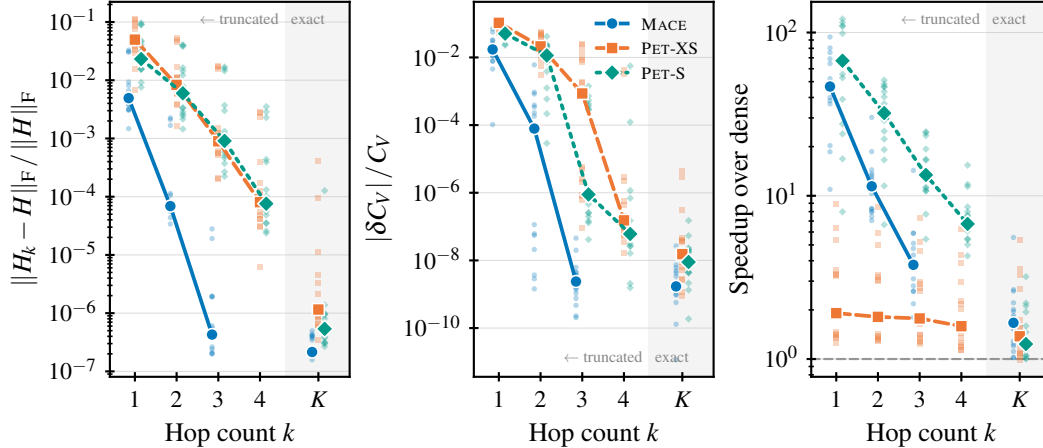


Figure 2: Truncation across the benchmark subset and the giant MOFs: relative Hessian error (left), relative C_V error at 300 K with the acoustic sum rule enforced (center), and end-to-end speedup over dense AD (right), versus hop count k ; each model’s exact K is set apart (shaded). Bold markers are medians over structures; small dots are individual structures. Sparse timings include pattern construction and coloring; the dashed grey line marks break-even.

Choosing 0.1 % as a threshold for C_V , for the vast majority of structures, MACE is converged at $k = 2$, PET-S at $k = 3$ and PET-XS at $k = 4$, with corresponding median speedups of $11\times$, $13\times$ and $1.6\times$. The exceptions are zeolites, as expected from the slower decay of the Hessian seen in Figure 3. The slowest-converging zeolite requires one more hop: $k = 3$ for MACE at $5.9\times$ speedup, $k = 4$ for PET-S at $5.4\times$, and the exact $k = K = 5$ for PET-XS, where little speedup remains. The modest speedups for PET-XS throughout reflect its high overall speed and small converged supercells—timings are dominated by fixed overheads and not HVPs. The errors in converged C_V are due to near-zero soft modes crossing the drop threshold of the pipeline (Appendix E); at exact K , C_V agrees with the dense reference to better than 1×10^{-5} relative. The Hessian error itself separates into the two contributions identified in Section 4; discarded couplings dominate until the pattern is nearly converged (Appendix H).

6 Conclusion

This work showed how to apply ASD to compute the Hessians of MLIPs. We explained how to obtain the sparsity pattern of MPNN-based MLIPs in closed form from their input graph by summing powers of the adjacency matrix. We then applied ASD to porous materials, whose large unit cells and porosity make them a natural best case for sparse Hessians. However, the interaction ranges of current foundation models are comparable to even these large unit cells, so ASD reproduces dense Hessians faithfully but at only modest speedups.

This motivated a truncated version of ASD. The exact sparsity pattern is the endpoint of a hierarchy of sparser patterns that only keep interactions between atoms separated by fewer graph hops. Using these sparser patterns neglects terms of the Hessian, but allows fewer colors, and hence fewer HVPs, for its materialization. We find that this approximation yields similar observables at a fraction of the computational cost. This somewhat surprising result is due to the decay of the Hessian with distance: truncating interactions yields a controlled loss of accuracy, which can be reduced by increasing the number of hops to be considered.

Several directions follow from here. Higher-order derivatives are the natural next step: three-phonon linewidths require contractions of the third-derivative tensor with a fixed direction, which reverse-mode AD evaluates without forming the tensor, at a small multiple of the Hessian’s cost [Gower and Gower, 2016]. This tensor inherits the sparsity of the Hessian, so sparsity patterns, colorings [Deussen and Naumann, 2019], and truncations of this work carry over directly; how it is best computed with higher-order and Taylor-mode techniques [Griewank and Walther, 2008, Griewank et al., 2000, Bettencourt et al., 2019] is left open. Truncation will not be as benign there as for heat capacity:

linewidths weight precisely the distant couplings that truncation discards, so the sufficient hop count will depend on the observable. A theoretical analysis of the truncation error, connecting it to the decay rates of Figure 3, would put the choice of hop count on firmer ground for all derivative orders.

Our results also expose message-passing depth as a performance-relevant design decision. The receptive field, so far chosen with accuracy in mind and little cost attached, directly sets the sparsity of higher-order derivatives, and keeping it compact now has a large payoff. For models whose interaction range is formally infinite, whether through deep receptive fields or explicit long-range terms, the Hessian is fully dense. Nevertheless, as physical interactions decay with distance, truncated ASD can still lead to large speedups where exact ASD cannot. Finally, since the predictive pipelines shown here are differentiable, at least in principle, they also pave the way towards fine-tuning on *experimental* instead of simulation data.

The presented truncated (or lossy) approach to ASD potentially also extends beyond the domain of MLIPs. For any derivative operator whose per-entry magnitude can be estimated ahead of time, truncated ASD materializes it at reduced cost by neglecting small, but nonzero, entries.

Code and data availability

All code and data of this work, including the JAX re-implementation of MACE, the experiment scripts, and the relaxed structures, Hessians, and derived observables of every experiment, will be made available in a single archive at <https://doi.org/10.5281/zenodo.00000000>, mirrored at <https://github.com/sirmarcel/tasd4mlip-archive>. A production-ready implementation of truncated ASD will be made available in `pet-jax` [Langer and Spies, 2026]. Further information and an overview of data sources can be found at <https://marcel.science/tasd4mlip>.

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A Benchmark structures

Benchmark set All heat-capacity benchmarks draw on the 233-structure porous-materials set assembled by Moosavi et al. [2022], comprising 214 MOFs, 9 COFs, and 9 zeolites (one further structure carries no class label), for which Gönninger et al. [2025] provide reference MACE heat capacities and phonon frequencies in unit cells and $2\times 2\times 2$ supercells. We take the published crystal structures as inputs and relax them within our own pipeline (Appendix E); no geometries are inherited from either upstream source.

Subset selection Computing converged Hessians for all 233 structures would be wasteful, so we benchmark on subsets, chosen deterministically as follows. Within each class, structures are ordered by farthest-point sampling [Kennard and Stone, 1969]: starting from the most central structure, the next pick is always the one most different from all previous ones. Distance is measured on three rank-transformed features: the logarithm of the unit-cell atom count, the number density, and the Hessian fill in the minimal exact supercell, which stand in for size, porosity, and the sparsity available to our method. The per-class orders are interleaved at 8 MOFs : 1 COF : 1 zeolite per ten ranks, so that subsets span all classes even though the full set is dominated by MOFs. A subset of size k is the first k entries of this ranking. The benchmarked twelve extend the first ten ranks: RSM1885 is skipped as infeasible in its MACE-converged supercell, and one further COF and two zeolites are added for class coverage.

Supercell selection Vibrational properties converge only once the simulation cell contains every interaction of the model, so supercells are chosen per structure and model. We build the model’s one-hop graph through its production input pipeline and propagate it to the Hessian hop count K (Section 4), tracking the periodic image that each interaction reaches. A supercell is admissible if no two of these interactions fold onto the same force-constant entry, which reduces to a divisibility test on image-offset differences. Among admissible cells, we pick the diagonal multiplier with the smallest volume; the search is exhaustive but finite, since the multiplier one larger than the largest offset difference in each direction is always admissible. Common practice instead interpolates the dynamical matrix in reciprocal space from a supercell judged large enough; we use exact cells to keep the pipeline simple and directly comparable to Gönninger et al. [2025]. Both the graph and K depend on the model, so the converged supercell, and with it N in Table 4, does too.

Giant MOFs For benchmarks beyond typical unit-cell sizes we use four MOFs with large primitive cells: MOF-177 (808 atoms), MOF-210 (1854), MIL-100 (2788), and MIL-101 (3604), taken unmodified from the structure library of RASPA2 [Dubbeland et al., 2016] and relaxed in our pipeline like all other structures. For MOF-177 and MIL-101, experimental heat capacities from calorimetry are available for comparison (Appendix I).

B Model implementations

We use two model families in this work: MACE-MP-0 medium [Batatia et al., 2022, 2025] (MACE) and PET-MAD-1.5 [Pozdnyakov and Ceriotti, 2023, Bigi et al., 2026, Malosso et al., 2026] in its XS and S variants (PET-XS, PET-S).

MACE combines equivariant message passing with higher-order Atomic Cluster Expansion features [Batatia et al., 2022]. The MP-0 medium foundation model has $L=2$ message-passing layers at a 6 Å cutoff, giving $K=4$, and is trained on Materials Project relaxation trajectories computed at the PBE(+ U) level of DFT [Batatia et al., 2025].

PET is an edge-to-edge transformer (within local neighborhoods) without built-in rotational equivariance [Pozdnyakov and Ceriotti, 2023]. The PET-MAD-1.5 models [Bigi et al., 2026] are pretrained on the OMat24 dataset [Barroso-Luque et al., 2026] and fine-tuned on the smaller, curated MAD dataset [Malosso et al., 2026], computed at the PBEsol level of DFT in a non-magnetic setting. The XS and S variants have $L=2$ and $L=3$ layers, giving $K=5$ and $K=7$. Within a fixed outer cutoff radius, PET shrinks its effective cutoff per atomic environment to bound the neighbor count. Since that cutoff depends on atomic positions, it contributes additional force terms, which we disregard: they couple atoms beyond the selected neighbor list and would decrease the very sparsity this work depends on (see the ablations in Appendix E).

Both models are distributed for PyTorch; to compute HVPs efficiently with JAX’s AD system, we rely on JAX re-implementations, using `pet-jax` [Langer and Spies, 2026] for PET and implementing MACE ourselves. We verify these on the first 50 structures of our benchmark ranking (Appendix A), comparing energies, forces and stresses in single and double precision against the upstream PyTorch models in double precision (Table 2). In double precision the re-implementations agree with upstream to far below the accuracy of the models themselves; in single precision, both stacks depart from that reference by the same amount, so the implementation contributes no error beyond rounding. One structure deviates further because a pair distance happens to fall where the two backends’ `tanh` kernels saturate differently, a property of the floating-point implementation rather than of either model. Allowing JAX to use TF32 for matrix multiplications (the default behavior on GPUs) is a different matter: it leads to errors nearly four orders of magnitude higher in the forces compared to full single precision, and is therefore not used in this work.

Table 2: Deviation of each prediction path from the upstream PyTorch model in double precision, over the first 50 structures of the benchmark ranking (Appendix A). Rows per model: upstream in single precision, then our JAX re-implementation in double precision, in full single precision (the production setting of this work), and with TF32 matrix multiplications. Cells are the median over the 50 structures with the worst structure in parentheses, energies per atom, forces and stresses as the largest deviating component. The first three rows are measured on CPU, the TF32 row on GPU. Running the full single-precision path on the GPU instead yields the same deviations to within rounding noise, so the last row isolates TF32, not the device.

Model	Prediction	ΔE (eV/atom)	ΔF (eV/Å)	$\Delta \sigma$ (eV/Å ³)
MACE	PyTorch fp32	3.3×10^{-6} (5.4×10^{-6})	4.0×10^{-5} (8.4×10^{-5})	5.5×10^{-8} (2.0×10^{-7})
	JAX fp64	1.2×10^{-8} (3.0×10^{-8})	3.6×10^{-7} (6.1×10^{-7})	5.9×10^{-9} (1.7×10^{-8})
	JAX fp32	1.0×10^{-7} (3.7×10^{-7})	4.0×10^{-5} (8.4×10^{-5})	5.5×10^{-8} (1.8×10^{-7})
	JAX fp32, TF32	8.3×10^{-5} (5.0×10^{-4})	2.9×10^{-1} (6.4×10^{-1})	4.5×10^{-4} (1.1×10^{-3})
PET-XS	PyTorch fp32	2.1×10^{-6} (7.1×10^{-6})	4.1×10^{-5} (9.0×10^{-5})	6.4×10^{-8} (2.2×10^{-7})
	JAX fp64	9.8×10^{-7} (2.1×10^{-6})	8.0×10^{-6} (2.3×10^{-5})	6.4×10^{-8} (2.0×10^{-7})
	JAX fp32	1.1×10^{-6} (2.3×10^{-6})	4.4×10^{-5} (5.6×10^{-4})	1.2×10^{-7} (7.1×10^{-7})
	JAX fp32, TF32	4.8×10^{-4} (1.0×10^{-3})	3.2×10^{-1} (6.5×10^{-1})	4.6×10^{-4} (1.2×10^{-3})
PET-S	PyTorch fp32	2.0×10^{-6} (7.2×10^{-6})	4.3×10^{-5} (8.8×10^{-5})	5.7×10^{-8} (1.9×10^{-7})
	JAX fp64	2.4×10^{-7} (4.6×10^{-7})	2.6×10^{-6} (4.8×10^{-6})	2.5×10^{-8} (8.0×10^{-8})
	JAX fp32	8.1×10^{-8} (2.8×10^{-7})	4.4×10^{-5} (8.6×10^{-5})	7.5×10^{-8} (2.3×10^{-7})
	JAX fp32, TF32	1.5×10^{-4} (6.0×10^{-4})	3.2×10^{-1} (6.7×10^{-1})	4.6×10^{-4} (1.2×10^{-3})

C Verifying the Hessian hop count

The two counts follow from the readout style: a node-readout energy term depends on the L -hop ball around its atom, of graph diameter $2L$, while an edge-readout term depends on the union of the L -hop balls around two adjacent atoms, of diameter $2L + 1$. We confirm them by measuring double-precision Hessians of small hand-built structures (chains, rings, chains with second-neighbor bonds, chains with pendant leaves) whose connectivity we control exactly and whose graph diameter exceeds the predicted K , taking the measured K as the largest graph distance at which any 3×3 Hessian block is nonzero. Freshly initialized models, whose random weights make every structurally allowed coupling generically nonzero, match the predicted count in every one of 96 measurements spanning $L = 1, \dots, 4$, both readout styles, all four graph families, and three seeds; the production checkpoints reproduce it at their trained depths, giving $K = 4$ for MACE-MP-0 medium, $K = 5$ for PET-XS, and $K = 7$ for PET-S. Beyond K , blocks are exactly zero. For PET, this is the hop count *after* the adaptive cutoff procedure, whose position dependence our sparse pipeline discards (ablated in Appendix E).

D Coloring on the atom graph

Two coordinates are coupled in the Hessian pattern if and only if their atoms are within K hops on the input graph. The coordinate sparsity pattern is therefore the $N \times N$ atom pattern with every nonzero replaced by a dense 3×3 block: sparsity is fully determined by pairs of atoms, and coordinates only

add a block structure. For this reason, it suffices to color the atom sparsity pattern, which has one ninth of the nonzeros, and then give coordinate $d \in \{0, 1, 2\}$ of atom i the color $3c_i + d$, where c_i is the atom’s color. By construction, this is a valid star coloring of the full coordinate pattern.

The timings we report combine two runs. Hessians and their HVPs come from a run that colored the coordinate pattern directly, the coloring time from a later run that colored the atom graph only. On all 234 sparse rungs of the benchmark the two colorings are identical, yielding the same HVPs, and only the cost of finding the coloring changes. At each model’s exact K , coloring the atom graph is 3 to $22\times$ faster than coloring the coordinate pattern, $17\times$ in the median, and over the whole benchmark the coloring stage drops from 22.8 h to 1.2 h.

E Computing heat capacity

Pipeline To compute heat capacity we follow the example of Gönheimer et al. [2025]: Initial unit cells are relaxed with BFGS over a `FrechetCellFilter` (L-BFGS with line search for the giant MOFs) to a maximum force of $5 \times 10^{-3} \text{ eV/\AA}$ in double precision, with MACE using the D3 correction and PET as-is. The Hessian is mass-weighted and diagonalized to give the frequencies entering Equation (1), zeroing imaginary frequencies and dropping near-zero modes with $|\nu| < 10^{-3} \text{ cm}^{-1}$. In comparisons between sparse and dense Hessians (Table 4 and Figure 2) we additionally enforce the acoustic sum rule, so that the acoustic modes cancel exactly instead of falling on either side of the drop threshold. We verify the pipeline by recomputing frequencies and heat capacities for all 233 structures in their unit cells with MACE, comparing against the published values of Gönheimer et al. [2025]. The two pipelines differ in the relaxed geometries and in the D3 Hessian, which is a finite difference over forces there and computed with AD here. The deviations are small: the largest frequency deviation within a structure is 0.7 cm^{-1} in the median over structures, and C_V at 300 K deviates by 3×10^{-5} relative in the median, with a worst case of 2.6%. Cells beyond $2 \times 2 \times 2$ were out of computational reach for Gönheimer et al. [2025].

Ablating precision, the D3 correction, and adaptive-cutoff forces Our pipeline departs from the reference Hessian calculations of Gönheimer et al. [2025] in three ways: MLIPs are evaluated in single rather than double precision, MACE Hessians omit the D3 dispersion correction, and PET Hessians disregard the force contributions of the adaptive cutoff procedure (Appendix B). To measure what each departure costs, we compute dense unit-cell Hessians for the ten-structure subset (Appendix A), flipping one axis at a time away from the reference configuration (MACE in double precision with D3, PET in double precision with adaptive-cutoff forces), and compare frequencies, Hessians, and heat capacities with the acoustic sum rule enforced. The single-precision ablation is run twice, with matrix multiplications in full single precision and in the TF32 matmul mode that JAX defaults to on GPUs (Appendix B). Table 3 summarizes the results. Single precision itself is harmless: with full matrix multiplications, heat capacities deviate by less than $5 \times 10^{-5} \%$ in the median. TF32 is not, and its effect is of the same size as omitting D3 or the adaptive-cutoff forces. All production Hessians in this work therefore pin matrix multiplications to full single precision, which makes each HVP 15–29 % slower for MACE and 55–97 % slower for PET than with TF32, at production supercell sizes on an H100. The production configuration, combining all three departures, deviates from the reference by at most 0.12 % in the median, with worst cases of a few percent. These worst cases are a threshold effect, not a shift of the spectrum: some structures have genuine low-lying optical modes near the drop threshold of the pipeline, and a perturbation of any origin can push one across it, adding or removing that mode’s $\sim k_B$ from C_V , a few percent in a small unit cell. The acoustic sum rule removes this ambiguity for the acoustic modes only. Every deviation above 1 % in the ablation is of this kind, with a single exception: for RSM0274, omitting D3 genuinely shifts the spectrum, by up to 67 cm^{-1} , for a 1.0 % change in C_V .

Table 3: Ablations of the production Hessian pipeline against the reference configuration of G nnheimer et al. [2025], computed as dense unit-cell Hessians for the ten-structure subset (Appendix A). Per ablation and model: median and worst relative deviation of C_V at 300 K with the acoustic sum rule enforced, worst single-mode frequency deviation, and median relative Frobenius error of the Hessian itself, each over the ten structures. The first two rows split single precision by matrix multiplication mode, full fp32 (the production setting) and TF32. The last row combines all departures and is the production configuration.

Ablation	Model	med. $ \delta C_V $ (%)	max $ \delta C_V $ (%)	max $ \Delta \nu $ (cm ⁻¹)	med. $\ \Delta \mathbf{H}\ _F / \ \mathbf{H}\ _F$
fp32 (full matmuls)	MACE	6.9×10^{-6}	3.6×10^{-5}	0.019	1.8×10^{-6}
	PET-XS	4.7×10^{-5}	2.4×10^{-2}	4.1	2.6×10^{-6}
	PET-S	3.3×10^{-5}	9.1×10^{-4}	0.14	2.1×10^{-6}
fp32 (TF32 matmuls)	MACE	0.010	3.0	51	5.3×10^{-3}
	PET-XS	0.029	3.0	112	4.9×10^{-3}
	PET-S	0.015	1.6	26	3.4×10^{-3}
no D3	MACE	0.056	2.9	67	3.6×10^{-3}
no adaptive-cutoff forces	PET-XS	0.12	2.5	377	7.0×10^{-3}
	PET-S	0.035	1.4	23	1.9×10^{-3}
fp32	MACE	0.056	2.9	67	3.6×10^{-3}
+ no D3	PET-XS	0.12	2.5	374	7.0×10^{-3}
+ no adaptive-cutoff forces	PET-S	0.035	1.4	23	1.9×10^{-3}

F Cost and accuracy across the benchmark

Table 4 lists wall times and heat-capacity errors at every hop count for every structure–model pair in the benchmark.

Table 4: Wall time and accuracy of dense, exact (K), and truncated ($k < K$) Hessians for every benchmark structure and model. N is the number of atoms in the supercell, chosen per model as the smallest multiple of the unit cell that accommodates its full interaction range (Appendix A); rows are grouped by chemistry class and sorted by the largest supercell across models within each group. Sparse wall times are end to end, including sparsity pattern construction and coloring as well as the Hessian evaluation itself. δC_V is the relative deviation of the heat capacity at 300 K from the same pair’s dense reference, computed as in Appendix E with the acoustic sum rule enforced; 0.00 marks magnitudes below 0.005%. MACE has $K=4$, so its $k=4$ entries appear in the K column.

Class	Structure	Model	N	Wall time (s)						δC_V (%)				
				dense	$k=1$	$k=2$	$k=3$	$k=4$	K	$k=1$	$k=2$	$k=3$	$k=4$	K
MOF	MOF-210	MACE	1854	555	51	69	109	–	213	–3.24	–0.04	0.00	–	0.00
		PET-XS	1854	111	34	35	37	37	42	–12.80	–5.10	–0.85	0.00	0.00
		PET-S	1854	379	48	57	70	88	260	–7.64	–1.41	0.00	0.00	0.00
	MIL-100	MACE	2788	1246	67	167	402	–	968	–0.77	0.00	0.00	–	0.00
		PET-XS	2788	227	36	38	44	52	63	–8.80	–0.56	0.00	0.00	0.00
		PET-S	2788	876	52	76	113	175	701	–2.48	0.00	0.00	0.00	0.00
	MIL-101	MACE	3604	3529	97	265	597	–	1327	–2.08	–0.02	0.00	–	0.00
		PET-XS	3604	316	35	37	43	50	59	–10.95	–1.51	0.00	0.00	0.00
		PET-S	3604	1248	52	75	112	166	643	–5.56	–0.02	0.00	0.00	0.00
	RSM1876	MACE	3969	4291	107	345	990	–	2788	–1.70	0.00	0.00	–	0.00
		PET-XS	1176	67	34	36	37	40	43	–8.93	–1.21	0.00	0.00	0.00
		PET-S	3969	1500	54	78	140	247	1328	–5.07	–1.19	0.00	0.00	0.00
	RSM0023	MACE	3840	4624	121	557	1769	–	4160	–1.19	–0.03	0.00	–	0.00
		PET-XS	768	45	33	35	35	39	45	–5.94	–0.86	–0.19	0.00	0.00
		PET-S	4800	2202	56	99	206	408	2198	–3.73	–0.06	–0.03	0.00	0.00
	MOF-177	MACE	6464	9431	117	312	659	–	1698	–2.63	0.00	0.00	–	0.00
		PET-XS	808	48	32	34	34	35	37	–12.68	–3.92	–0.72	0.00	0.00
		PET-S	6464	3900	58	84	167	253	1220	–6.62	–0.74	0.00	0.00	0.00
	RSM1877	MACE	6615	23 285	490	2788	10 680	–	22 759	–1.15	–0.02	0.00	–	0.00
		PET-XS	1764	116	35	38	42	53	75	–13.80	–2.78	0.00	0.00	0.00
		PET-S	6615	4699	65	145	340	742	4458	–4.98	–1.10	0.00	0.00	0.00
	RSM1162	MACE	2916	1137	56	143	439	–	948	–2.40	–0.08	0.00	–	0.00
		PET-XS	864	45	31	33	33	37	40	–13.51	–1.54	0.00	0.00	0.00
		PET-S	6912	4439	62	110	230	519	2782	–4.44	–2.13	0.00	0.00	0.00
	RSM0010	MACE	7700	39 619	839	5622	22 015	–	38 950	–0.01	–0.01	0.00	–	0.00
		PET-XS	2112	113	35	37	43	56	78	–8.39	–1.26	0.00	0.00	0.00
		PET-S	8448	6845	69	187	481	1031	6552	–7.05	–0.39	–0.05	0.00	0.00
	RSM0274	MACE	7840	33 275	515	3220	11 851	–	26 961	–0.45	0.00	0.00	–	0.00
		PET-XS	2016	124	35	38	43	55	71	–7.79	–0.31	–0.17	0.00	0.00
		PET-S	8960	8723	76	187	522	1094	7283	–4.04	–0.13	–0.04	0.00	0.00
	RSM0047	MACE	3888	3022	68	209	636	–	1503	–1.76	0.00	0.00	–	0.00
		PET-XS	1152	64	34	36	39	44	53	–5.38	–1.30	0.00	0.00	0.00
		PET-S	9216	6967	63	136	283	593	3291	–7.45	–2.15	–0.03	0.00	0.00
COF	20561N3	MACE	7680	21 529	285	1678	5410	–	11 001	–0.79	0.00	0.00	–	0.00
		PET-XS	960	48	34	34	34	38	45	–10.07	–2.71	0.00	0.00	0.00
		PET-S	5760	3091	59	107	242	454	2372	–5.51	–0.78	0.00	0.00	0.00
	18150N2	MACE	5850	13 406	257	1168	3666	–	8754	–1.10	0.00	0.00	–	0.00
		PET-XS	3120	200	37	40	44	51	61	–11.14	–3.53	–1.04	0.00	0.00
		PET-S	9360	9426	78	195	388	827	4280	–3.90	–2.41	0.00	0.00	0.00
zeolite	VFI	MACE	4860	7759	168	681	2116	–	4212	–3.20	–0.09	0.00	–	0.00
		PET-XS	1296	64	34	36	37	42	49	–12.31	–4.67	–1.55	–0.33	0.00
		PET-S	5832	3657	66	147	324	671	3616	–2.31	–1.91	–0.14	0.00	0.00
	NPT	MACE	6912	11 453	122	614	1946	–	5182	–7.03	–0.60	0.00	–	0.00
		PET-XS	864	45	33	34	36	38	41	–10.15	–5.25	–2.36	0.00	0.00
		PET-S	6912	4428	66	139	340	687	3656	–5.07	–4.18	0.04	–0.56	0.00
	AFI	MACE	6912	16 600	240	1214	4274	–	9249	–5.60	–0.01	0.00	–	0.00
		PET-XS	864	44	36	34	35	39	43	–12.04	–5.63	–2.48	–0.42	0.00
		PET-S	9216	8079	74	186	489	1063	6618	–6.10	–2.38	–0.07	–0.01	0.00

G Force-constant decay

Figure 3 is computed from the dense Hessians of the benchmark subset and the giant MOFs in their model-converged supercells (Table 4), plus RSM1885 for the PET models; its MACE-converged supercell is infeasible (Appendix A). Only rows belonging to atoms in the original unit cell appear, as replica atoms are equivalent and carry the same force constants. The hop count of a pair (i, j) is the smallest k for which $(\mathbf{A}^k)_{ij}$ is nonzero, with $\mathbf{A}$ the input-graph adjacency of Section 4; $k=0$ labels the on-site blocks, and $\|\Phi_{ij}\|$ is the Frobenius norm of the pair’s 3×3 force-constant block. Quantiles are taken over the pooled pairs of each chemistry class, with the giant MOFs counted as MOFs. The same Hessians confirm the sparsity pattern at production scale: beyond each model’s K , every block is exactly zero.

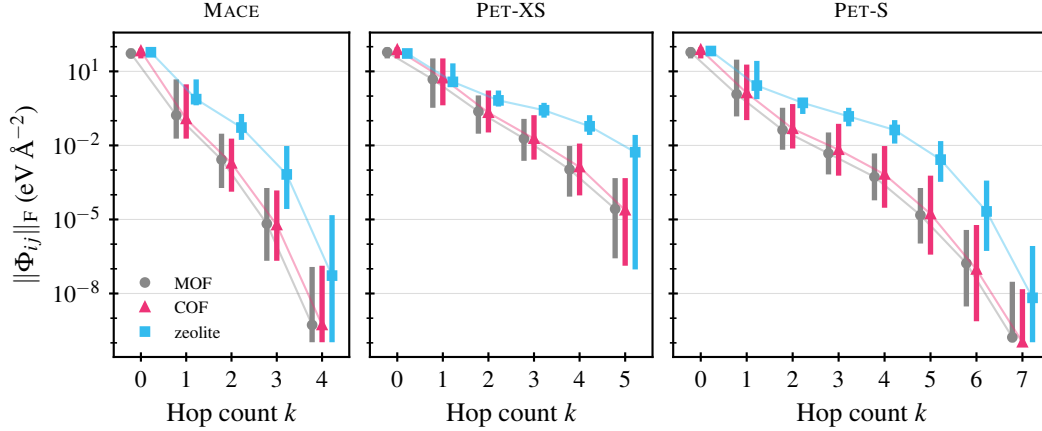


Figure 3: Force-constant block norms $\|\Phi_{ij}\|$ versus hop count k , split by chemistry class, one panel per model: median and 10–90% band over all coupled pairs of the benchmark subset and the giant MOFs, plus RSM1885 for the PET models.

H Truncation error decomposition

A truncated Hessian is not the dense Hessian with distant couplings set to zero: star coloring prevents collisions only within the assumed sparsity pattern, so neglected couplings contribute to retained entries (Section 4). The error therefore has two parts: the *discarded* couplings themselves, and the *contamination* of the retained entries. No entry appears in both, so their squared Frobenius norms add up to the squared total error. We compute the split from the stored Hessians of every sparse pattern in Table 4, plus RSM1885 for the PET models as in Appendix G; the Hessians are single precision, while the comparison is done in double precision. Figure 4 plots one against the other: contamination tracks the discarded couplings at a roughly constant fraction, a median of 0.60 across four orders of magnitude, until it bottoms out at the single-precision noise floor measured by the exact- K patterns, where floating-point noise, not truncation, sets the error.

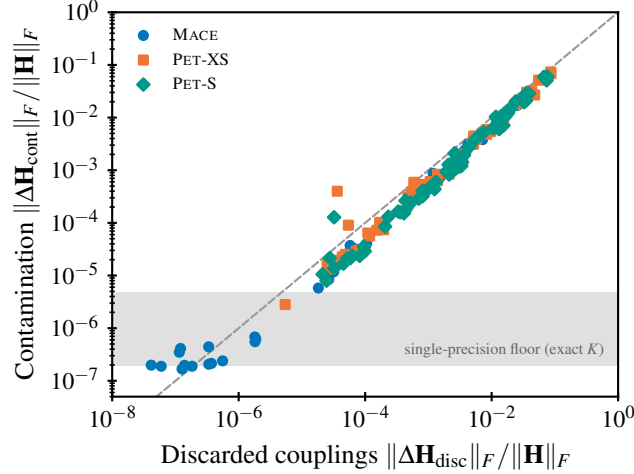


Figure 4: Contamination of the retained entries against the discarded couplings, both relative to the dense Hessian norm: one marker per (structure, model, hop count) for every truncated pattern of Table 4 and of RSM1885 for the PET models. The dashed line is equality; the band spans the central 80% of the exact- K errors, the single-precision noise floor.

I Comparison with experiment

For the two giant MOFs with published calorimetric heat capacities [Kloutse et al., 2015, Liu et al., 2017], Figure 5 compares predictions from dense Hessians in converged supercells against the measured curves; since $C_p \approx C_V$ for these solids in this temperature range, the comparison is direct. All models overestimate: at 300 K, MACE lies 30% above experiment for MOF-177 and the PET models around 17%, consistent with the systematic over-softening of predicted phonon spectra observed by Gönner et al. [2025]. For MIL-101 the gap widens to 84% and 61–64%, too large to attribute to the harmonic approximation or the potentials alone; the measured sample, with defects and possible residual guest species, likely differs substantially from the ideal crystal we compute. Discrepancies of this kind are precisely what large-scale Hessian calculations make visible, and what fine-tuning on experimental data would correct.

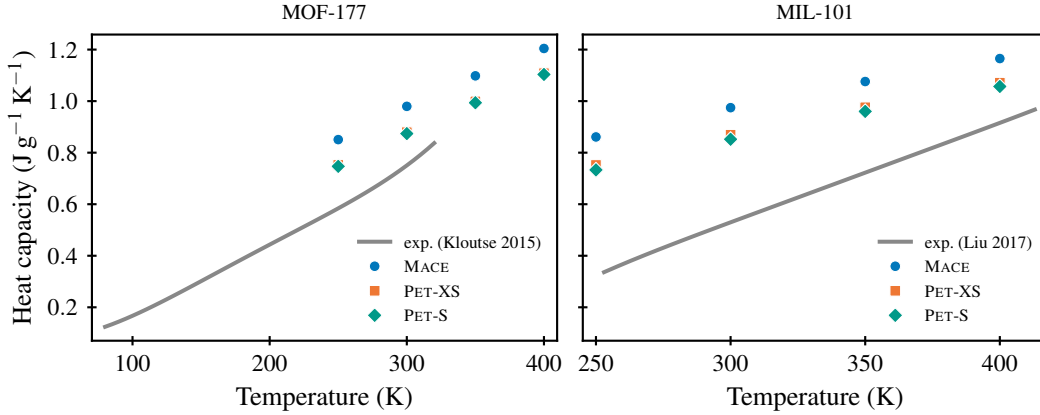


Figure 5: Predicted heat capacity against calorimetry for the two giant MOFs: measured $C_p(T)$ curves (grey; MOF-177 from Kloutse et al., 2015, MIL-101 from Liu et al., 2017) and C_V from dense MLIP Hessians in converged supercells (markers), with the acoustic sum rule enforced.